Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Methanamide (CHONH2)
Basis set : cc-pVDZ
Charge : 0
Multiplicity : 1
Total e- : 24
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -168.96056339
E1 (Ha) : -168.72195263
Gap (Ha) : 0.23861076
HF (Ha) : -168.94623212

CONVEX-ROOF METHODS

Methods run : gcg
Restarts : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 9
Frozen e- : 18
Active e- : 6
Active orbs : 5
Spin-orbitals : 10
Hilbert dim : 100

SPIN PROPERTIES

<S2> : 0.000000
S : 0.0000
<Sz> : 0.000000
Multiplicity : 1
Spin state : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.988948
<D> / site : 0.597790

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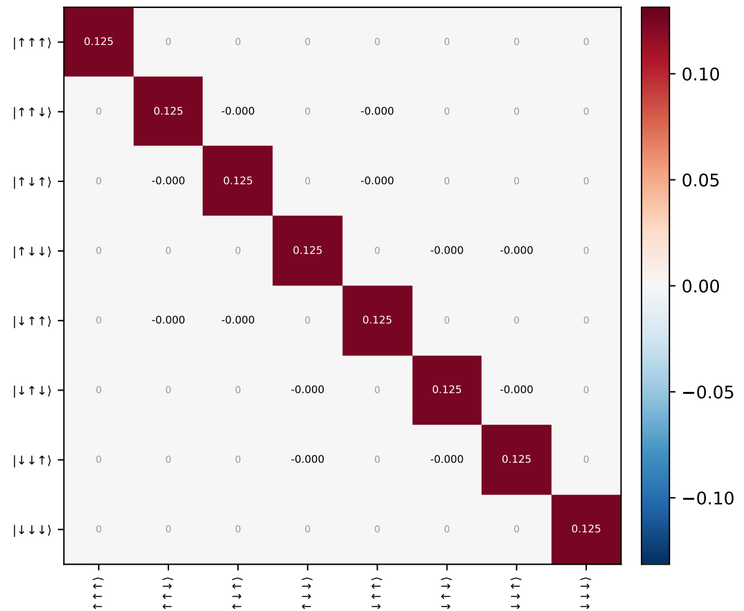
Body order : 3
Clusters : 10

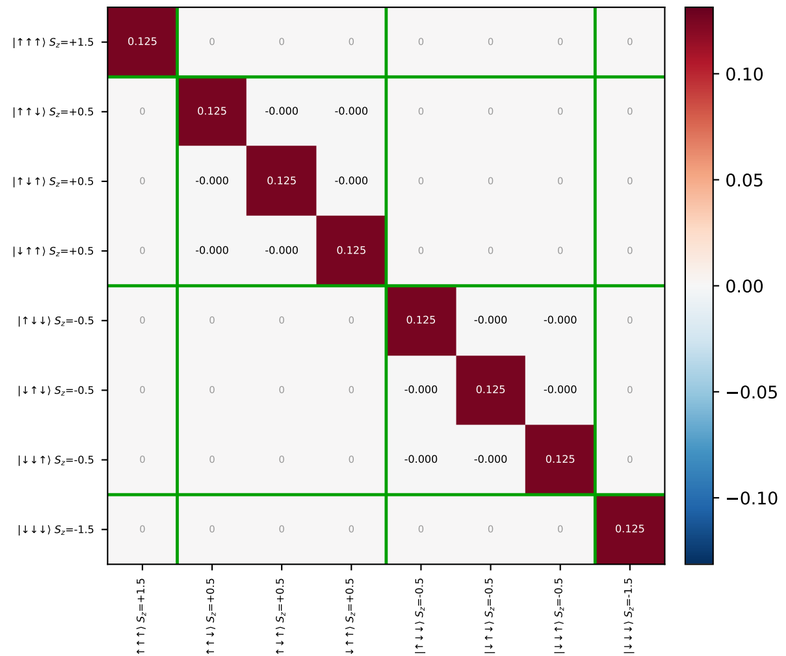
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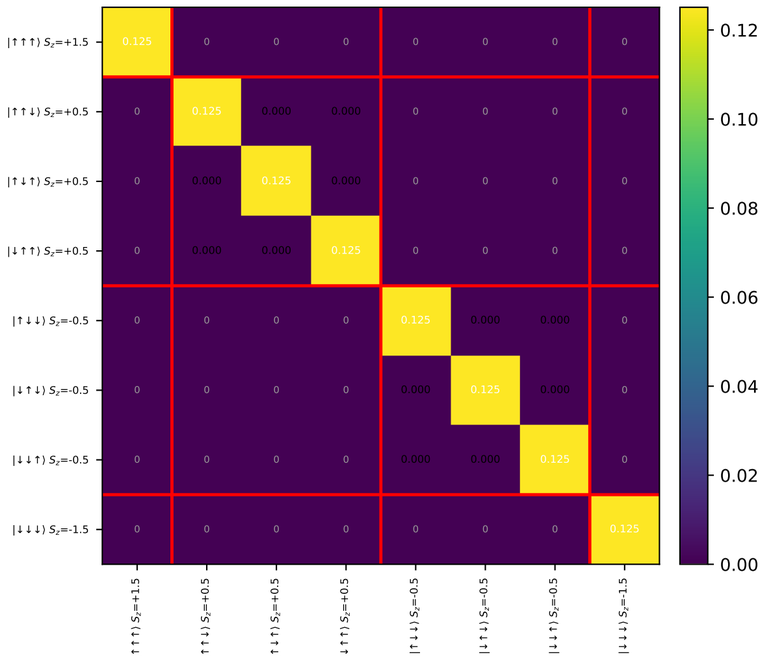
Summary — 3-Body RDMs (Methanamide, CAS(6,5))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2	0.125000	2.079441	—	0.000000	0.000223	0.439460	STRONGLY MIXED
0,1,3	0.178470	1.893960	—	0.000000	0.000744	0.278703	STRONGLY MIXED
0,1,4	0.125597	2.077109	—	0.000000	0.001770	0.443768	STRONGLY MIXED
0,2,3	0.131047	2.053712	—	0.000000	0.000115	0.469508	STRONGLY MIXED
0,2,4	0.157300	1.947854	—	0.000000	0.000014	0.438372	STRONGLY MIXED
0,3,4	0.272269	1.593283	—	0.000000	0.003801	0.168539	MIXED
1,2,3	0.148948	1.994699	—	0.000000	0.001164	0.443601	STRONGLY MIXED
1,2,4	0.126533	2.073525	—	0.000000	0.001309	0.440829	STRONGLY MIXED
1,3,4	0.394889	1.092477	—	0.000000	0.017243	0.346206	MIXED
2,3,4	0.377417	1.178193	—	0.000000	0.000854	0.372188	MIXED

3-Body RDM (Sites 0,1,2) — same matrix in all three views



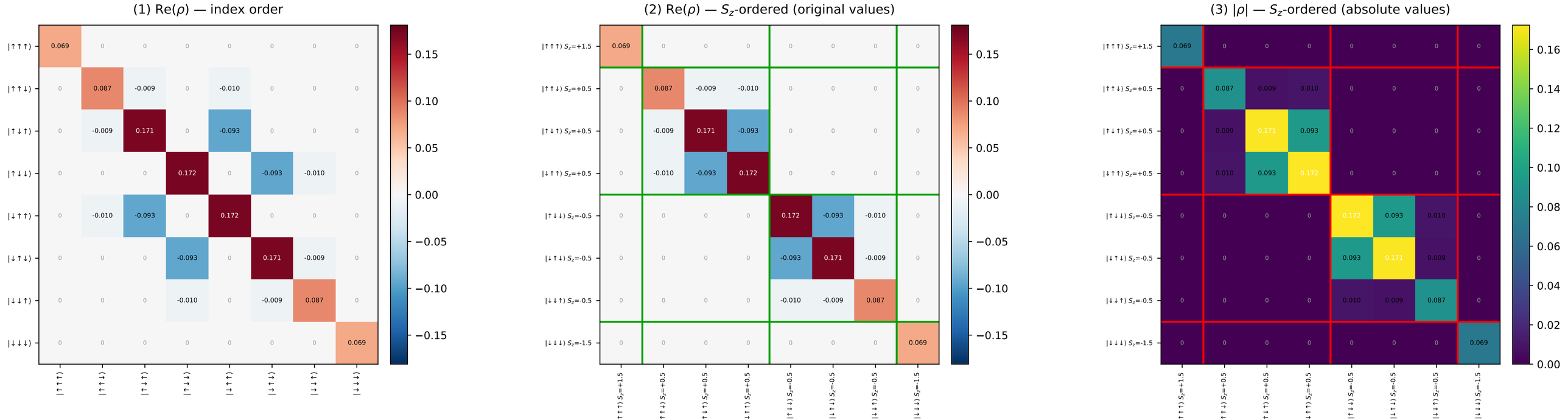




Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000223 ≤ C_{UB} = 0.439460 verdict: GME

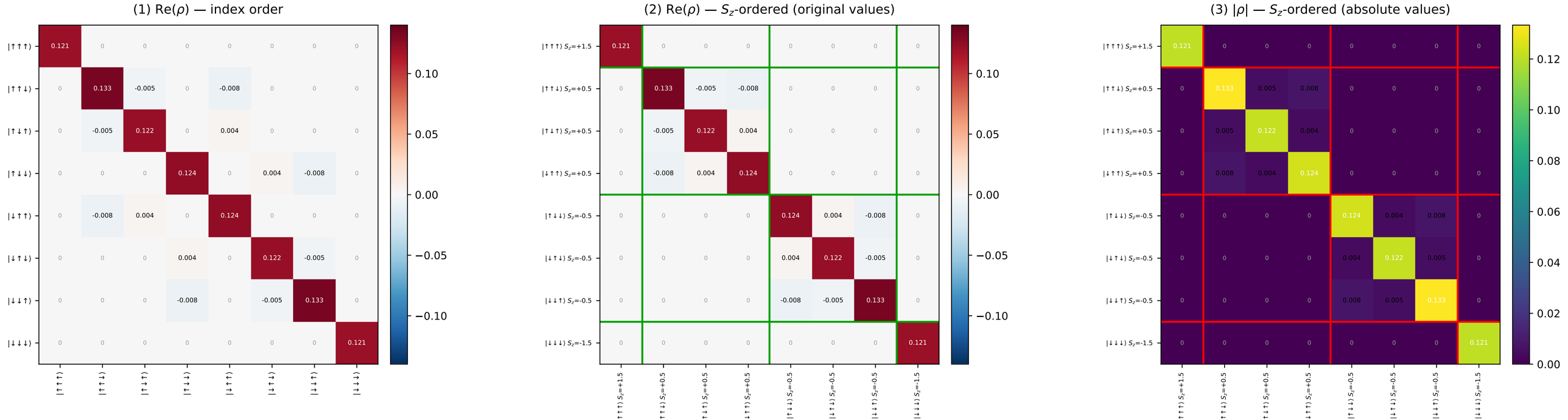
3-Body RDM (Sites 0,1,3) — same matrix in all three views



Trace=1.000000 Purity=0.178470 Entropy=1.893960 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000744 ≤ C_{UB} = 0.278703 verdict: GME

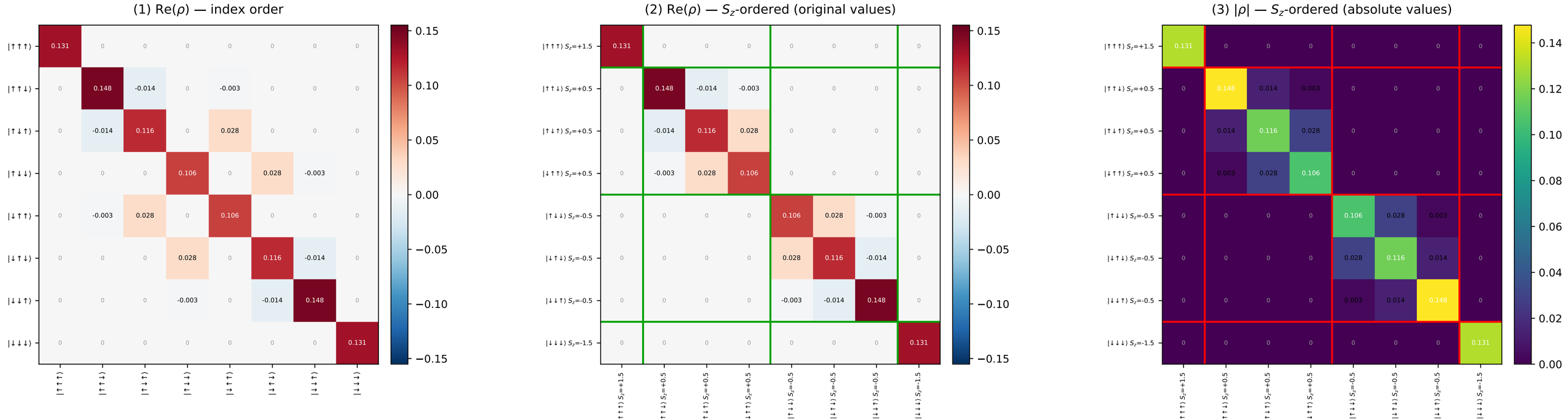
3-Body RDM (Sites 0,1,4) — same matrix in all three views



Trace=1.000000 Purity=0.125597 Entropy=2.077109 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001770 ≤ C_{UB} = 0.443768 verdict: GME

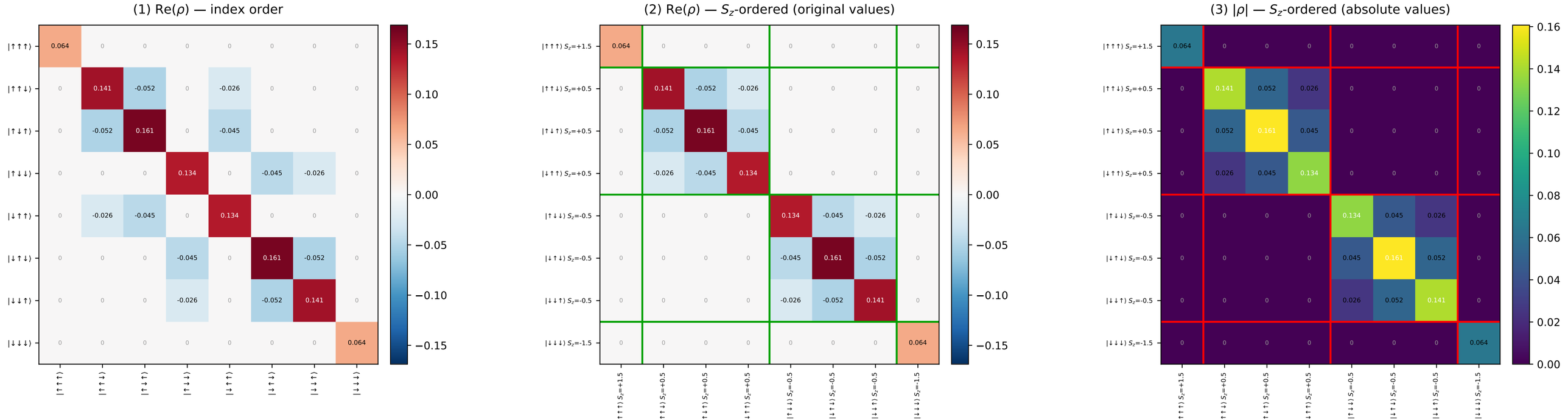
3-Body RDM (Sites 0,2,3) — same matrix in all three views



Trace=1.000000 Purity=0.131047 Entropy=2.053712 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000115 ≤ C_{UB} = 0.469508 verdict: GME

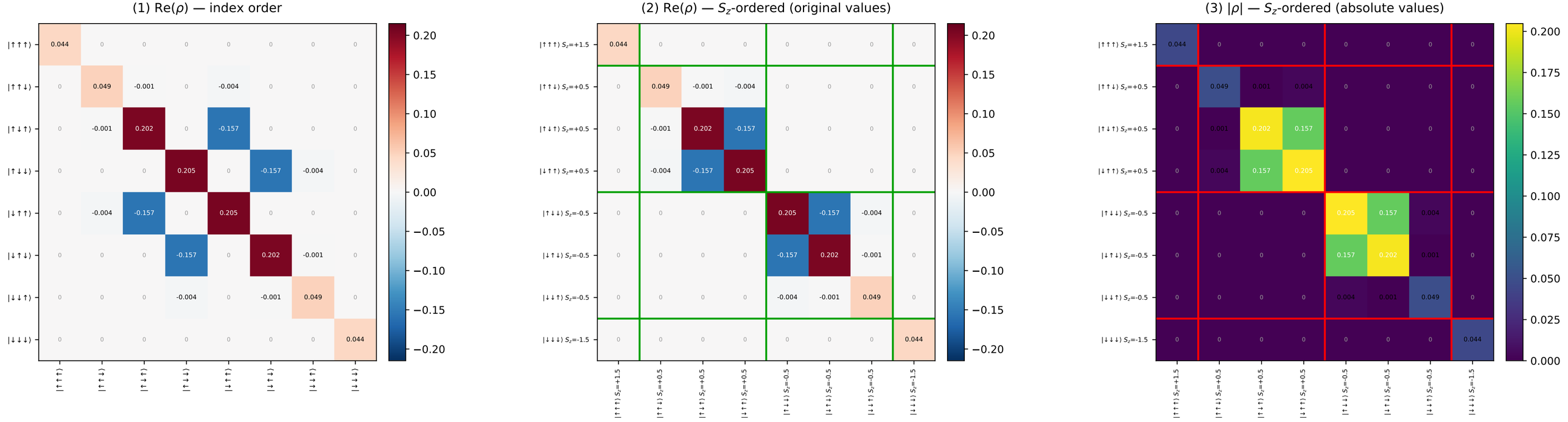
3-Body RDM (Sites 0,2,4) — same matrix in all three views



Trace=1.000000 Purity=0.157300 Entropy=1.947854 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000014 ≤ C_{UB} = 0.438372 verdict: GME

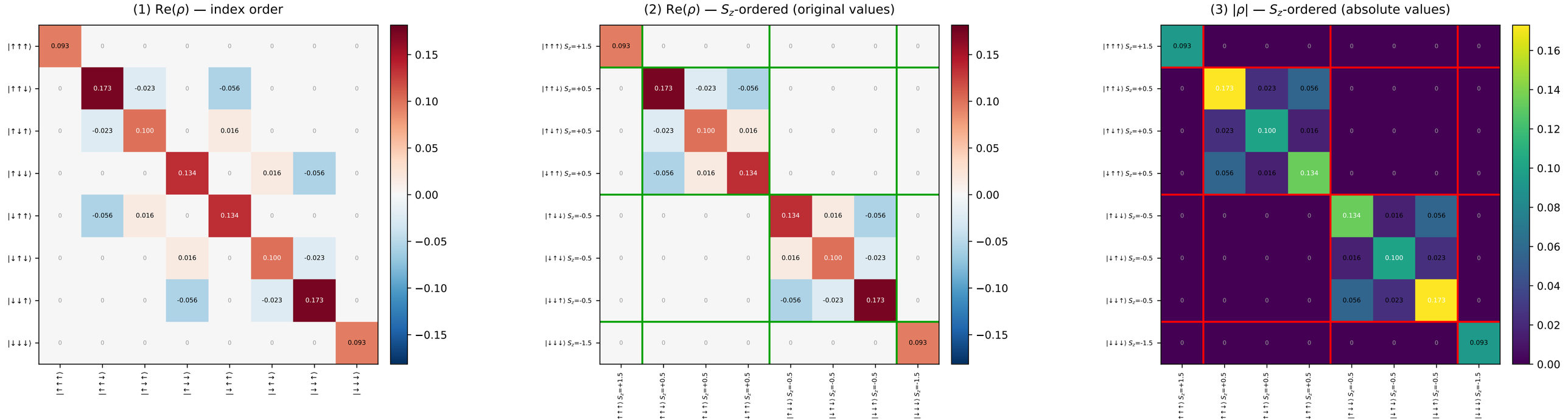
3-Body RDM (Sites 0,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.272269 Entropy=1.593383 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.003801 ≤ C_{UB} = 0.168539 verdict: GME

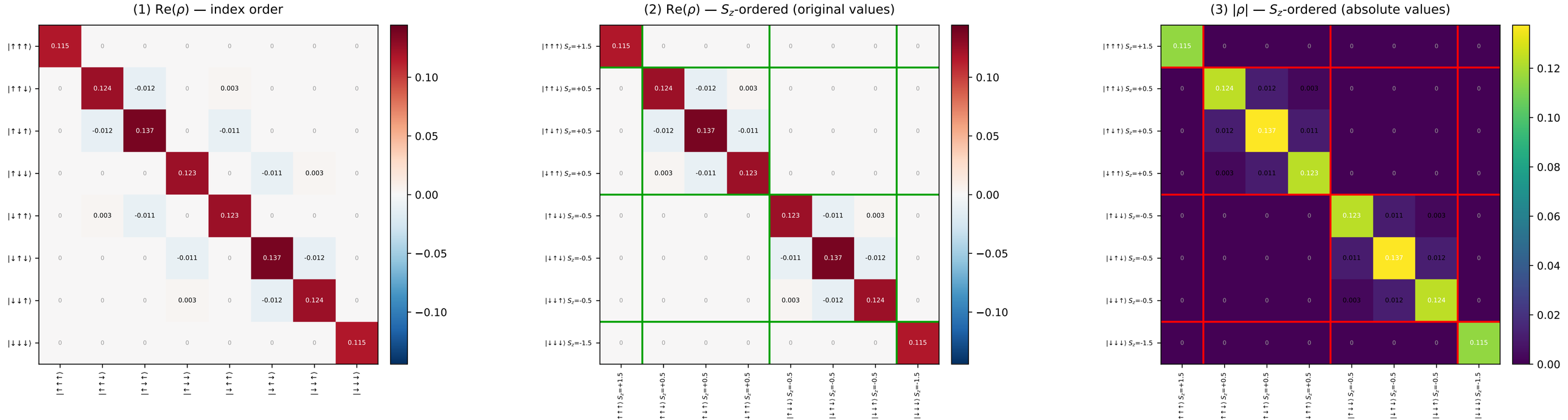
3-Body RDM (Sites 1,2,3) — same matrix in all three views



Trace=1.000000 Purity=0.148948 Entropy=1.994699 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001164 ≤ C_{UB} = 0.443601 verdict: GME

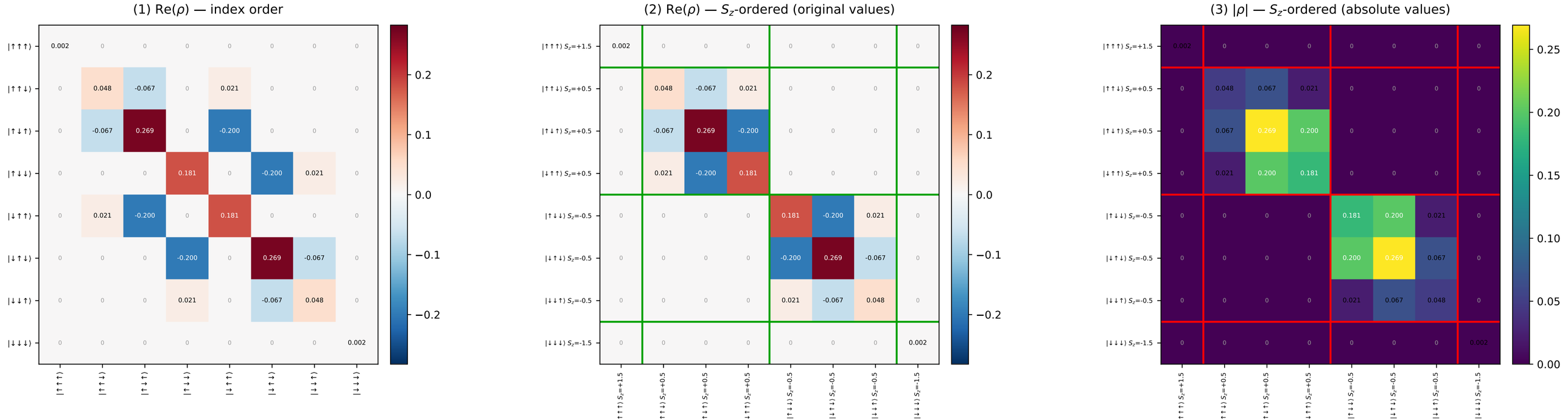
3-Body RDM (Sites 1,2,4) — same matrix in all three views



Trace=1.000000 Purity=0.126533 Entropy=2.073525 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001309 ≤ C_{UB} = 0.440829 verdict: GME

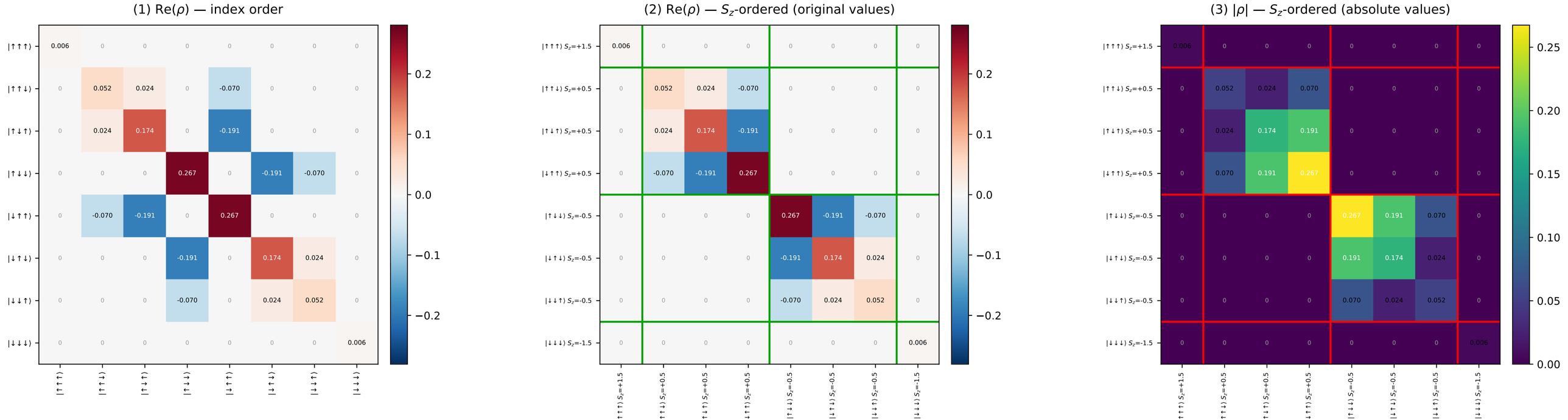
3-Body RDM (Sites 1,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.394889 Entropy=1.092477 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.017243 ≤ C_{UB} = 0.346206 verdict: GME

3-Body RDM (Sites 2,3,4) — same matrix in all three views



Trace=1.000000 Purity=0.377417 Entropy=1.178193 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000854 ≤ C_{UB} = 0.372188 verdict: GME